Hospital & Clinic Management Platform — Product Requirements

Nov 4, 2026 · @Sumit

Version 1.0 — final. This document is the frozen product requirement for Finmeyo's hospital and clinic management platform: 26 functional modules plus enterprise platform requirements, all in Release 1.0. Any change from here is recorded in the change log at the end.

Overview

Finmeyo will build a cloud-based (SaaS) platform that runs an entire hospital or clinic in one place: front desk, OPD, IPD, lab, pharmacy, billing, insurance/TPA, HR, payroll, accounts, ambulance, website and marketing. Customers pay a subscription and install nothing.

Target customers

- **Large hospitals:** 15+ doctors, multiple branches, full IPD, OT, ICU, emergency and insurance/TPA operations. The product is designed for this customer first.
- **Small and mid-size clinics:** 3–15 doctors, usually one branch, OPD with lab and pharmacy. They use the same product with unneeded modules switched off.

Goals

- Shorter front-desk waiting, with a token and queue system that keeps lines moving
- Doctors see a patient's full history on one screen and write a prescription in under a minute
- Every rupee is accounted for: consultations, tests, medicines, beds, insurance claims and doctor payouts
- Owners see daily revenue and department performance on one dashboard
- Patients receive bookings, reports and bills on their phone and WhatsApp

Scope: the whole hospital in one system

- Everything from staff attendance and payroll to patient care, billing, TPA, doctor payouts and accounts runs in this platform
- The hospital needs no other software; a Tally/Zoho export is offered only for the convenience of its CA

Product decisions

The product is built **complete, to large-hospital standards**, and smaller clinics configure it down by switching modules off. Release 1.0 ships whole: OPD, IPD, TPA, all major specialties, website builder, lead pages and patient app from day one.

Topic	Decision	Impact on the product
Target customer	Designed for large hospitals; small clinics via configuration	Every module is built; a clinic switches off what it does not need (e.g. IPD, OT, blood bank)
Coverage	OPD and IPD, all major specialties	26 specialty templates in Release 1.0 (see Specialty coverage)
Insurance / TPA	Mandatory in Release 1.0	Full flow from pre-authorization to settlement, including NHCX
Market	India now; other countries later	GST, ABDM and NHCX for India; tax, currency and coding designed so other countries can be added
Pricing	Per clinic for now; may move to per login, per doctor or per bed	Configurable pricing engine, so the model changes without code changes
Language	English only for now; multilingual later	All interface text kept in resource files from day one, so adding a language is translation only
Go-to-market	Start with a few pilot clinics and hospitals	Pilots begin only when Release 1.0 is complete; pilot feedback sets the price
Quality bar	Complete product, nothing half-built	Every module tested, with data migration and training material ready before launch
Company	Finmeyo	All registrations (Razorpay, ABDM, NHCX, WhatsApp, DLT) in Finmeyo's name
Payments	Razorpay	Razorpay at launch; connector design allows other gateways later
First region	Delhi NCR	Delhi, Noida, Gurugram, Ghaziabad, Faridabad

Topic	Decision	Impact on the product
Pilot terms	Free trial for a period	Trial length still to be decided
Hosting	Hostinger VPS	Shared SaaS on Hostinger VPS at launch; dedicated, private cloud and on-premise options for enterprise customers

The app name and the per-clinic fee will be decided later.

Users and roles

Each user sees only what their job needs. Admins can create new roles and change permissions without code.

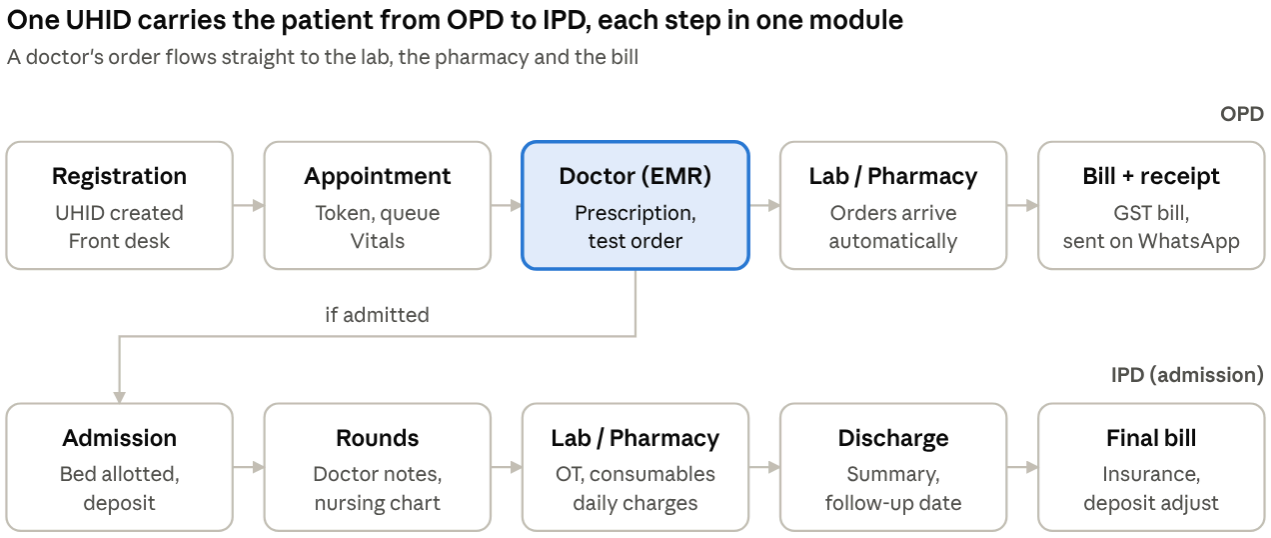
Role	Can do	Cannot see
Super admin (owner)	Everything: branches, staff, rates, reports, subscription	—
Branch admin	Own branch's staff, rates and reports	Other branches' data
Doctor	Own patients, EMR, prescriptions, test orders, own earnings	Other doctors' earnings, full billing
Nurse	Vitals, nursing notes, medication chart, bed status	Billing, reports
Receptionist	Registration, appointments, queue, OPD billing	Clinical note details
Lab technician	Lab orders, samples, result entry	Billing, prescriptions
Pharmacist	View prescriptions, dispense, stock, pharmacy billing	Clinical notes
Accountant	Billing, payments, refunds, GST reports, doctor payouts	Clinical notes
TPA desk	Policies, pre-auth, claims, settlements	Clinical notes beyond claim documents

Role	Can do	Cannot see
HR manager	Employees, roster, attendance, leave, payroll	Clinical and billing data
Finance manager	Accounts, expenses, vendor payments, financial reports	Clinical notes
Department head	Own department's roster, leave approvals, reports	Other departments' HR data
Ambulance coordinator / crew	Dispatch, tracking, trip records	Clinical notes beyond trip details
Patient (app)	Own bookings, reports, bills, prescriptions	Anyone else's data
Call centre agent / supervisor	Calls, bookings, leads, complaints; supervisors also monitor calls and agent performance	Clinical notes, billing beyond dues
CRM manager / field executive	Segments, campaigns, referral doctor network, field visits	Clinical notes
Store manager / purchase officer	Items, indents, purchase orders, GRN, issues, stock audits, vendors	Clinical notes, patient billing
Sub-store in-charge (OT/ICU/ward)	Own store's stock, indents, patient-wise consumption	Other stores' stock values, purchasing
Quality manager / medico-legal officer	Incidents, RCA, escalations, legal cases, legal holds, case bundles, disputes	POSH cases (Internal Committee only)
Infection control nurse / compliance officer	Infection surveillance, antibiotic stewardship, BMW records, statutory registers, licence calendar	Billing, HR data
Radiologist / imaging technician	Scan schedule, worklists, image acquisition, reporting queue	Billing beyond own studies
Field staff (home care, sample collection, delivery)	Assigned visits and deliveries, vitals, samples, payment collection	Records of patients not assigned to them

Every sensitive action (bill cancellation, discount, record change) is written to an audit log: who, when, and what changed.

Patient journey

A patient moves from front desk to bill under a single UHID, and from OPD straight into IPD when admitted. Every module below serves this journey.



patient journey · OPD and IPD

No paper and no re-entry anywhere: a doctor's order flows automatically to the lab, the pharmacy and the bill.

Functional requirements by module

All 26 modules ship in Release 1.0.

#	Module	Covers	Build
1	Patient registration	UHID, search, family linking, ABHA	1
2	Appointments and queue	Schedules, booking channels, tokens, displays	1
3	OPD / EMR	History, vitals, specialty templates, e-prescription, orders	1
4	Lab and radiology	Worklists, results, analyzers, partner labs, B2B lab business	2
5	Pharmacy	Dispensing, pharmacy billing	2

#	Module	Covers	Build
6	Billing and GST	Combined bills, GST logic, packages, payments	1
7	Insurance and TPA	Empanelment to settlement and disbursement	2
8	Doctor payouts	Engagement types, rules engine, statements, TDS	2
9	IPD	Beds, admissions, nursing, OT, discharge	2
10	Reports and dashboards	Operational reports and exports	1
11	Multi-branch	Branches, shared UHID, branch rates and stock	1
12	Website builder	Templates, drag-and-drop, connected blocks, SEO	3
13	Lead generation pages	Campaign landing pages, lead tracking	3
14	Hospital-grade modules	Emergency, ICU, day care, dialysis, OT, blood bank, MRD, CSSD and more	2
15	HR and payroll	Employees, rosters, attendance, leave, statutory payroll	2
16	Finance and accounts	Automatic entries, payables, receivables, P&L, GST/TDS	2
17	Ambulance management	Fleet, dispatch, live tracking, billing	2
18	CRM and call centre	Telephony, agent console, campaigns, referral network, complaints	3
19	Inventory and supply chain	Stores, procure-to-consume, controls, assets, own serial numbers	2
20	Incidents, escalation and medico-legal	Incidents, RCA, legal cases, legal hold, disputes	2
21	Clinical safety and nursing	CPOE, drug alerts, eMAR, early warning, nursing workflow	2
22	Infection control and compliance	HICC, antibiotic stewardship, BMW, PCPNDT, licence calendar	2
23	KPIs, MIS and analytics	Hospital KPIs, dashboards, report builder	3

#	Module	Covers	Build
24	Check-ups, home care, delivery	Check-up routing, home services, pharmacy delivery	3
25	Radiology information system	Scheduling, worklists, reporting queue, PACS	2
26	Facility services	Maintenance, mortuary, canteen, parking, DMS	3

1. Patient registration

- New patient: name, age, gender, mobile, address, emergency contact, allergies
- A unique **UHID** per patient, valid across all branches and departments
- Search existing patients by mobile number, with a duplicate-patient warning
- Family linking (several family members on one mobile number)
- Link an ABHA ID

2. Appointments and queue

- Doctor schedules: days, hours, slot length, leave, visiting consultants
- Three booking channels: walk-in, phone (via front desk), online (patient app, website, WhatsApp)
- Department- and doctor-wise **token/queue**, shown on waiting-area TV displays
- Reschedule, cancel, mark no-show
- Referrals from one doctor to another department

3. OPD / EMR (doctor workspace)

- One screen with the patient's full history: visits, diagnoses, reports, allergies, current medicines
- Vitals: BP, pulse, temperature, weight, SpO2 (entered by nurse, front desk or doctor)
- **Specialty templates** for every major specialty (see Specialty coverage)
- Diagnosis with ICD-10 code search
- **E-prescription:** medicine search, dose, duration, advice, and the doctor's favourite sets
- Print or send the prescription on WhatsApp
- Order lab and radiology tests from the same screen
- Follow-up date and reminder; patient education videos and articles (condition, procedure, diet) sent on WhatsApp from a content library

4. Lab (pathology) and radiology

- Doctor orders land directly in the lab worklist
- Sample collection, barcode labels, sample status
- Test master: test, parameters, normal ranges by age and gender, rate
- Result entry with abnormal values highlighted, pathologist approval
- Report PDF on the hospital letterhead, sent to the patient by WhatsApp or app
- Radiology scheduling, worklists and reporting in module 25 (RIS), with PACS integration
- Lab analyzer interfacing (HL7/ASTM) so results flow in without typing
- Outsourced tests managed end to end (see Outsourced and partner labs below)

Outsourced and partner labs

Many tests (specialised panels, genetics, histopathology, rare assays) go to outside labs. The hospital needs to manage these partners commercially and operationally, so every outsourced test is tracked from order to payment and the margin on it is known.

Tie-ups and commercials

- Partner lab register: name, NABL/CAP accreditation, agreement copy, validity, contacts, pickup schedule, payment terms
- Test-wise commercials per partner: partner rate (cost to the hospital), hospital MRP (price to the patient), margin per test; flat rates, discount on partner MRP, or revenue share
- Multiple partners per test, with a preferred partner and rules to pick by cost, TAT or accreditation
- Rate history and renewal alerts at 60/30/7 days before agreement expiry

Routing rules

- Per test: in-house only, outsourced only, or in-house with fallback to a partner
- Temporary outsourcing when an analyzer is down or a reagent is out of stock, with automatic switch-back
- Patient bills always at the hospital's rate, whatever the routing

Sample dispatch and tracking

- Dispatch batches: samples, partner, courier/pickup person, time, with a printable or digital manifest
- Partner barcode printed alongside the hospital barcode; cold-chain temperature recorded where needed
- Status: collected → dispatched → received by partner → in process → report received

- TAT tracking against the agreed TAT, with alerts on delays and rejected samples (recollection triggered)

Reports from partners

- Results received by API/HL7 integration where the partner supports it, otherwise by portal download or PDF upload
- Discrete results stored where available, so trends and alerts work like in-house tests
- Report issued in the partner's format or on the hospital letterhead (as the agreement allows), released after internal verification

Billing, reconciliation and payment

- Partner's monthly statement auto-generated from dispatch records and matched to the partner's invoice; duplicates and missing tests flagged
- Payment with TDS, and accounting entries posted automatically (module 16)
- Revenue-share settlements calculated where applicable

Reports

- Partner-wise volumes, cost, revenue, margin and TAT performance
- Most-outsourced tests, to show which tests are worth bringing in-house

B2B lab business (hospital lab as a reference lab)

- Referring clinics, nursing homes and collection centres as B2B clients, each with its own rate list and credit terms
- Sample pickup scheduling and field staff tracking (module 24)
- Client portal for ordering, sample status and report download
- Monthly client invoices, outstanding tracking and payment collection
- Client-wise volumes, revenue and receivables reports

The same partner model applies to outsourced imaging (e.g. PET-CT sent to a partner centre) in module 25.

5. Pharmacy and inventory

- Prescriptions appear directly on the pharmacy screen
- Medicine master: brand, salt, manufacturer, HSN code, GST rate
- Stock by batch, expiry, MRP and purchase rate
- Purchase entry, suppliers, purchase returns
- Low-stock and expiry alerts
- Pharmacy billing as a GST tax invoice

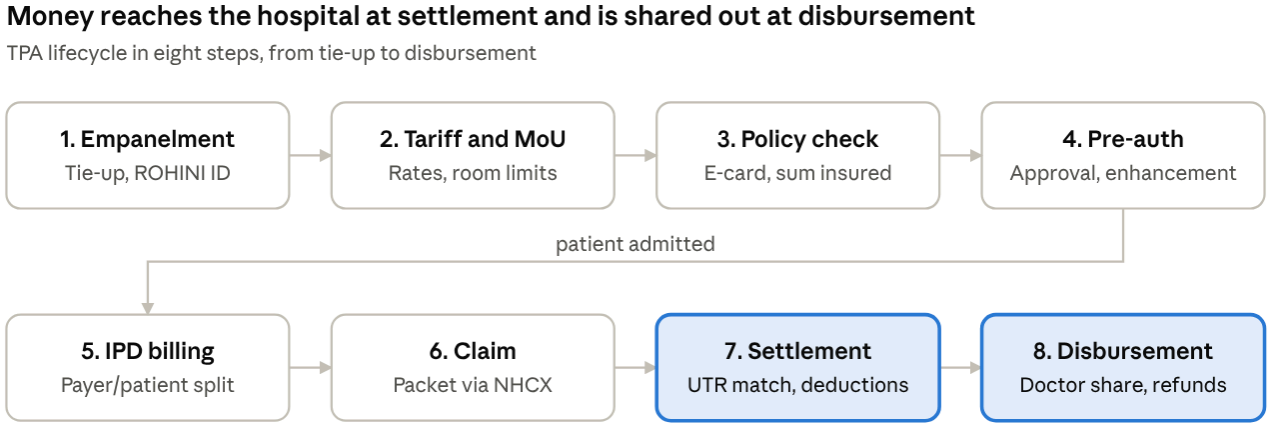
- Stock, purchasing and all non-pharmacy stores are handled in module 19 (Inventory and supply chain)

6. Billing and GST

- Service master: rates per service and department, doctor-wise consultation fees
- One **combined bill** per visit: consultation, tests, procedures, medicines
- **GST logic:** exempt services (consultations, diagnostics, treatment) get a Bill of Supply; taxable items (medicines, cosmetic procedures, high-tariff rooms) get a Tax Invoice; a mixed bill shows both separately
- GST rates and HSN/SAC codes editable from the admin panel, never hard-coded
- Health check-up packages
- Discounts (with approval), refunds, advances and deposits
- Payment by cash, UPI, card, or several modes on one bill
- Daily cash closing and shift-wise collection reports
- GSTR-1 / GSTR-3B exports; e-invoicing for B2B bills when turnover exceeds ₹5 crore

7. Insurance and TPA

Insurance and TPA are a core part of Release 1.0. A large share of a hospital's IPD revenue comes through insurance, and rejected or delayed claims are where most money gets stuck.



TPA lifecycle · 8 steps, tie-up to disbursement

Steps 1–2 happen once per payer; steps 3–8 repeat for every insured patient.

Empanelment (tie-up) management

- Register of every insurer, TPA, government scheme and corporate the hospital is empanelled with, or applying to

- Empanelment status: identified → applied → documents requested → inspection → tariff negotiation → agreement signed → active → suspended / expired
- Hospital credentials stored once and reused in every application: ROHINI ID, NABH/NABL certificates, Clinical Establishment registration, fire and biomedical waste licences, bed count, specialties, equipment list, key doctors
- PPN (Preferred Provider Network) participation and package agreements with public-sector insurers
- Agreement copy, validity dates, signatories and contact persons per payer
- Renewal alerts 60/30/7 days before expiry; expired agreements block new cashless admissions for that payer with a warning
- Tariff negotiation history: proposed vs agreed rates, version-controlled rate lists

Payers and setup

- Payer types: insurance companies, TPAs, government schemes (PM-JAY/Ayushman Bharat, CGHS, ECHS, state schemes), corporates, and cash
- Each payer's MoU: tariff, package rates, room-category limits, discounts, payment terms, validity
- Several rates per service (cash, TPA, scheme package); the bill picks the right rate for the payer automatically
- Payer-wise document checklists

Patient policy

- Policy number, TPA, member ID, sum insured, remaining balance, policy dates, co-pay %, room-rent limit
- Upload of e-card and ID scans
- Multiple policies per patient (primary and secondary)

Pre-authorization (cashless approval)

- Pre-auth form before or at admission, with diagnosis (ICD-10), procedure, estimated cost and documents
- Planned and emergency admissions
- Status tracking: sent → query received → answered → approved (amount) / rejected
- **Enhancement** requests during treatment
- Reminders and deadline alerts for TPA queries

Impact on billing

- Live view of approved amount, running bill, payer share and patient share
- Automatic handling of non-payable items and proportional deductions for room-rent limits

- Co-pay and deductible calculation
- Final bill split into payer share and patient share

Discharge and claim

- Final approval request with discharge summary and bills
- Claim packet: all documents in one PDF, ordered per the payer's checklist
- Claim submission through NHCX where the payer is connected; otherwise a ready packet for the TPA portal or email
- Document set for patients filing reimbursement claims

Settlement and reconciliation

- Claim status: submitted → query → approved → settled / partially settled / rejected
- Payment advice capture (upload, email parse or NHCX response): claim-wise amount paid, deductions with reasons, TDS
- **Bank reconciliation:** match each payer's bank credit (UTR) to the claims it covers; one payment often covers many claims, so the system splits and matches it
- Unmatched receipts and unpaid claims flagged for follow-up
- **Deduction handling:** classify every deduction as recoverable from the patient, appealable, or a write-off, with approval; appeals tracked to outcome (module 20)
- TDS certificates (Form 16A) collected per payer and matched to TDS deducted
- Ageing of pending claims (0–30, 31–60, 61–90, 90+ days) with escalation to the payer
- Payer-wise outstanding, deduction trends and reconciliation reports

Disbursement (sharing the money out)

- **Doctor share:** for TPA cases, the doctor's share is released only after settlement, adjusted pro-rata for deductions where the doctor's rule says so (module 8)
- **Patient refunds:** if the payer settles more than expected and the patient paid an advance or co-pay, the refund is calculated and paid back with approval
- **Patient recovery:** deductions marked recoverable create a patient due, with a payment link sent by WhatsApp
- **Accounting:** settlements, deductions, write-offs and TDS post to accounts automatically (module 16)

TPA desk dashboard

- Today's admissions needing pre-auth, pending TPA queries with deadlines, enhancements due
- Patients ready for discharge awaiting final approval
- Claims to submit, claims awaiting payment, and amount outstanding per payer

- Agreements expiring soon

Government schemes

- PM-JAY/Ayushman Bharat: package-based billing, beneficiary verification, data for the scheme portal
- CGHS/ECHS: billing as per their rates and referrals

Reports

- Payer-wise revenue, approval rate, average settlement time, deductions, pending claims

8. Doctor payouts

Hospitals engage doctors on many different terms. The system stores each doctor's rule and calculates their share from every bill automatically.

Engagement types

- **Salaried:** fixed salary (through HR payroll) plus incentives
- **Revenue share / fee-for-service:** % or fixed amount per consultation, procedure, surgery or IPD visit
- **Retainer plus share:** fixed monthly amount plus a share above a target
- **Visiting consultant:** amount per visit or per case
- **Package/surgery share:** separate shares for surgeon, assistant and anaesthetist
- **External referring doctors:** referral commission, configurable within the law and NMC rules

Rules engine

- Different % by service, department and payer (e.g. 60% for cash patients, 50% for TPA)
- When to count: on billing, or on payment received (after settlement for TPA cases)
- Configurable effect of discounts and deductions on the doctor's share
- Minimum guarantees and target-based slabs
- Adjustment of the doctor's share when a TPA deducts from a claim

Payout process

- Statement per cycle (usually monthly): every case, amount, share, deductions
- Live earnings and statements in the doctor app
- Doctor queries and disputes with resolution tracking
- Payout after approval, with TDS (Section 194J) deducted, and a bank transfer file
- Automatic accounting entries for payouts (module 16)

Reports

- Doctor-wise revenue and payout, each doctor's profit contribution, pending payouts

9. IPD (admissions and beds)

- Ward, room and bed setup, with type (general, semi-private, private, ICU) and rate
- Live bed board: vacant, occupied, under cleaning
- Admission: doctor, bed, deposit, insurance
- Bed transfers
- Doctors' daily round notes and orders
- **Nursing:** vitals chart, medication administration record (when and by whom), intake/output
- Daily charges posted automatically: room, nursing, doctor visits
- **OT:** surgery booking, surgeon/anaesthetist/team, OT notes, consumables
- Discharge summary from templates, final bill, deposit adjustment

10. Reports and dashboards

- Owner dashboard: today's patients, today's revenue, bed occupancy
- Revenue by department and by doctor
- Most-ordered tests and best-selling medicines
- Pending payments and insurance claims
- Excel and PDF export

11. Multi-branch

- One hospital group with many branches
- Patient data shared across branches under the same UHID
- Separate rates, staff and stock per branch
- Group-level and branch-level reports

12. Website builder (templates and drag-and-drop)

Every hospital or clinic builds its own website in 15–30 minutes without a developer. The website is connected to the platform: doctors, departments, timings and rates come from the masters, and online bookings land directly in the appointment module. This is the key difference from generic builders such as Wix or WordPress.

Templates

- 10–15 ready templates: multi-specialty hospital, small clinic, dental, eye, IVF/gynaecology, skin/cosmetic, diagnostic lab, physiotherapy
- Choosing a template fills in the clinic's name, logo, address, doctors and services automatically
- Templates can be switched any time without losing content
- Theme: colours, fonts, button style, light/dark

How the specialty is decided (all by configuration)

No website is fixed to one specialty. What the clinic selects at sign-up decides the website's type and content:

1. **Clinic type:** single-specialty clinic, multi-specialty clinic, or hospital (with IPD)
2. **Specialties:** one or many (dental, eye, gynaecology, orthopaedics, dermatology, cardiology…), from the same department master used across the platform
3. **Template suggestions:** every template is tagged (e.g. "dental", "multi-specialty", "hospital"); matching templates are shown first, but any template can be chosen
4. **Automatic content:** a page per selected specialty with its doctors, services and packages, plus specialty-specific sample text and images (e.g. "root canal, braces, implants" for dental)
5. **Later changes:** adding a specialty (e.g. ENT) adds its page and doctors automatically; switching one off hides its page

Clinic type	Website shape
Single-specialty (e.g. dental)	Focused on one specialty: treatments, before/after gallery, doctor profile, booking
Multi-specialty clinic	Department menu, a page per department, full doctor list with filters
Hospital	All of the above plus emergency number, IPD and facilities, insurance partners, health packages, branches

Drag-and-drop editor

- Drag blocks onto a page, reorder and remove them
- Per-block settings: text, image, colour, spacing, background
- Desktop, tablet and mobile preview; show or hide blocks per device
- Undo/redo, autosave, draft and publish
- Pages: Home, About, Departments, Doctors, Services, Health packages, Blog, Gallery, Careers, Contact

- Menu (navigation) and footer editor

Connected blocks (live platform data)

Block	What it shows	Data source
Book appointment	Choose department, doctor and slot, then book	Appointment module (live slots)
Doctor list / profile	Photo, qualifications, specialty, timings, fee	Doctor master
Departments	A page per department with its doctors	Department master
Health packages	Package, included tests, price, "Book now"	Package master
Report download	Patient fetches a report with mobile/UHID and OTP	Lab module
Online payment	Pay a bill or advance	Billing + Razorpay
Teleconsultation	Book an online consultation	Video integration
Patient reviews	Selected ratings from feedback	Feedback module
Bed availability (optional)	Vacant beds, only if the hospital wants it	IPD bed board

Standard blocks are also available: hero banner, text, image, gallery, video, FAQ, testimonials, contact form, Google Map, WhatsApp chat button, stats counter, insurance partner logos, blog list.

Domain and hosting

- Free subdomain on Finmeyo's domain
- Custom domain (e.g. `www.cityhospital.in`) with free SSL (HTTPS) issued automatically
- Hosting and CDN handled by the platform; nothing for the clinic to manage
- Multi-branch: one website for all branches, or a page per branch

SEO and marketing

- Per-page title, description, URL and social share image
- Sitemap, Google Search Console, Google Analytics and Meta Pixel
- Google Business Profile link and structured data (doctor, hospital) for better search visibility

- Blog editor for health articles
- Contact-form and booking leads collected in one place for front-desk follow-up

Rules and safety

- Website language: English for now; multilingual later
- Privacy policy and terms templates; cookie consent
- Reminders about medical advertising rules (NMC guidelines), e.g. no "guaranteed cure" claims
- Patient data from forms protected under the DPDP Act

Release and pricing: Release 1.0 (Build 3). The pricing engine can sell it as an add-on or as part of a plan.

13. Lead generation pages and lead management

For every promotion (health camp, package offer, new doctor, cosmetic offer) the hospital can create a dedicated landing page and track every lead it generates until that lead becomes a patient. The website builder's editor and blocks are reused here.

Landing page templates by purpose

Purpose	Page content	Form fields
Health check-up package offer	Package, included tests, offer price, countdown	Name, mobile, preferred date
Free health / screening camp	Camp date, venue, tests offered, register button	Name, mobile, age, area
Specialty consultation (IVF, knee, spine, cardiac)	Condition, treatment, doctor profile, success stories	Name, mobile, condition, city
Surgery cost estimate	Procedure, insurance/EMI options, "Get an estimate"	Procedure, insured or not, mobile
Cosmetic / dental offer	Before/after gallery, offer, limited slots	Name, mobile, treatment
New doctor / department launch	Doctor introduction, video, introductory offer	Name, mobile, preferred time
Second opinion	Report upload, specialist opinion	Name, mobile, report file
Corporate health tie-up	Packages for companies	Company, contact person, employee count

Purpose	Page content	Form fields
Vaccination / seasonal drive	Vaccine, date, price	Name, mobile, child's age
Careers (staff hiring)	Open positions	Name, mobile, role, resume

Page settings

- A unique URL per campaign (e.g. `cityhospital.in/knee-camp`) and a QR code for offline posters
- Start and end dates; after expiry the page shows "offer ended" or redirects
- Coupon/offer codes that apply in billing
- Form builder: fields, required/optional, OTP verification of mobile numbers to cut fake leads
- After submission: thank-you page, automatic WhatsApp reply, or direct slot booking
- A/B testing: two versions of a page, the better converter wins
- Consent checkbox (DPDP Act) and permission to call or WhatsApp

Lead sources (tracking)

- UTM tracking: Google Ads, Facebook/Instagram, YouTube, WhatsApp, SMS, poster QR, referral
- Meta Lead Ads and Google lead forms flow directly into the system
- Conversions sent back to Google Ads and Meta Pixel to improve ad performance

Lead management (full CRM and call centre in module 18)

- All leads in one place: website, landing pages, phone calls, walk-in enquiries, WhatsApp
- Stages: New → Contacted → Interested → Appointment booked → Visited → Converted / Lost (with reason)
- Assignment to callers/counsellors (automatic or manual) with follow-up reminders
- Call notes, click-to-call, WhatsApp templates
- Leads matching an existing patient's mobile link to that record (no duplicates)
- One-click appointment from a lead

Reports (where the money comes from)

- Per campaign: leads, appointments, patients, revenue
- Per source: cost per lead and ROI (ad spend entered manually or pulled from ad accounts)
- Conversion by counsellor

- Best-performing landing pages

Release and pricing: Release 1.0 (Build 3), alongside the website builder. The pricing engine can sell it as a "Marketing add-on".

14. Hospital-grade modules

Large hospitals need these modules beyond OPD and IPD. Each one can be switched off, so small clinics simply keep them disabled.

Module	What it does
Emergency / casualty	Quick registration (name can be added later), triage colour codes, emergency billing, transfer to OPD/IPD, MLC (medico-legal case) register and police intimation record
ICU / critical care	ICU bed board, hourly vitals chart, ventilator and infusion records, ICU scores, critical alerts
Day care	Chemotherapy, dialysis, minor procedures: same-day admission and discharge, session-based billing
Dialysis unit	Machine-wise scheduling, session records, recurring patient cycles
OT management	OT schedule and calendar, WHO surgical safety checklist, anaesthesia record, implants and consumables, surgeon/anaesthetist fees
Blood bank / blood storage	Donor registration, grouping, cross-match, stock and expiry, issue records (as per licence rules)
Medical records (MRD)	Case-file records, file issue/return, ICD coding, retention periods
Births and deaths	Records and certificates, data for government registration
Consent management	Procedure-wise consent forms, digital signature/thumbprint, video consent where required
CSSD	Sterilization cycles for instrument sets, issue/return, batch tracking
Diet / kitchen	Doctor diet orders, ward-wise diet lists, patient meal records

Module	What it does
Housekeeping and linen	Bed-cleaning requests after discharge, linen issue and laundry
Ambulance	Full detail in module 17 (Ambulance management)
Biomedical and assets	Equipment maintenance and breakdowns; the full asset register is in module 19
Store and purchase	Full detail in module 19 (Inventory and supply chain)
Staff, roster and attendance	Full detail in module 15 (HR and payroll)
Quality and NABH	Automatic NABH quality indicators (e.g. infection rate, readmissions, waiting time) and audit checklists; incidents in module 20
Feedback and complaints	Patient feedback (OPD/IPD), complaint tickets and resolution
Visitors and bed management	Visitor passes, live bed availability, bed transfer requests

15. HR and payroll

Every employee (doctors, nurses, technicians, housekeeping, security, admin) is managed here, from joining to salary to exit. Hospitals run 24x7, so shifts, night duty and on-call are the most important part of this module.

Employee records

- Employee master: personal details, photo, department, designation, grade, reporting manager, branch, joining date, employment type (permanent, contract, consultant, trainee, outsourced)
- Documents: degrees, registrations (NMC/State Medical Council/Nursing Council), licence expiry, ID proofs, offer letter, with **alerts before licence expiry** (needed for NABH)
- Bank account, PAN, Aadhaar (masked), UAN (PF), ESIC number
- Training and certifications (BLS/ACLS, fire safety, infection control) with expiry tracking
- Organization chart

Recruitment and onboarding

- Job openings (linked to the website careers page), applications, interview scheduling and feedback
- Offer letters from templates, with digital acceptance
- Onboarding checklist: documents, login creation, uniform, ID card, induction training

Shifts, roster and attendance

- Shift types: morning, evening, night, general, split, on-call; ward- and department-wise rosters
- Weekly/monthly roster creation, shift-swap requests and approvals
- Minimum staffing rules with alerts when a ward is short of nurses
- Attendance via biometric/face devices, mobile app (GPS geo-fence), or manual entry with approval
- Automatic calculation of late arrivals, early exits, overtime and night-shift allowances
- Attendance and billing for outsourced (agency) staff

Leave

- Configurable leave types and policies (CL, SL, EL, maternity, comp-off) with carry-forward and encashment rules
- Leave applied and approved in the app, reflected in the roster automatically
- Branch-wise holiday calendars

Payroll

- Configurable salary structures: basic, HRA, allowances (night, on-call, overtime), deductions
- Salary computed automatically from attendance, leave and overtime
- **Statutory compliance:** PF, ESI, Professional Tax, TDS (Form 16), LWF, with reports for challans and returns
- Salary advances and loans with EMI deductions
- Arrears, bonus, increments, full and final settlement
- Payslips by app or email, and bank transfer files in the bank's format
- Payments to contract staff and consultants, with TDS

Performance and exit

- Appraisal cycles with goals/KPIs (e.g. patient feedback for nurses, OPD volume for doctors)
- Exit: resignation, notice period, clearance checklist (assets, dues), experience letter

Employee self-service (app)

- Mark attendance, view roster, apply for leave, download payslips, update documents, read announcements

HR reports

- Headcount, attrition, attendance, overtime cost, department-wise salary cost, licence expiry list, statutory reports

16. Finance and accounts

Every billing, pharmacy, purchase, payroll and doctor-payout transaction becomes an accounting entry automatically. Owners see daily cash and monthly profit and loss without waiting for their CA.

- **Chart of accounts:** ready-made for hospitals (revenue by department, expenses by category), and editable
- **Automatic entries:** OPD/IPD bills, pharmacy sales, purchases, salaries, doctor payouts, TPA settlements, refunds
- **Expenses:** utilities, rent, maintenance and other expenses, with bill photos and approval
- **Vendor payments:** supplier bills, due dates, payments, TDS, outstanding
- **Receivables:** patient dues, TPA/insurance outstanding, corporate credit
- **Banking:** bank accounts, statement import, bank reconciliation, UPI/card settlement matching
- **Petty cash** and branch-wise cash books
- **Reports:** day book, ledger, trial balance, profit and loss, balance sheet, cash flow; profitability by department and branch
- **GST:** GSTR-1, GSTR-3B, input tax credit; **TDS** return data
- **Budgets:** department-wise budget versus actual
- Financial-year closing and period locking (past entries cannot be changed)
- Tally/Zoho export for the CA

17. Ambulance management

Covers the hospital's own and partner ambulances end to end: from the incoming call to the patient reaching emergency, to billing and vehicle maintenance. In an emergency every minute counts, so response time is this module's headline measure.

Fleet

- Ambulance master: registration number, type (BLS, ALS, ICU-on-wheels, neonatal, patient transport, hearse), branch, owner (hospital or partner/vendor)

- Documents with expiry alerts: RC, insurance, fitness, permit, PUC, ambulance registration
- Per-vehicle equipment checklist: oxygen cylinder, ventilator, monitor, defibrillator, stretcher, emergency drugs
- Live status: available, on trip, under maintenance, off duty

Crew

- Duty rosters for drivers, paramedics/EMTs, nurses and doctors (for ALS), linked to the HR roster (module 15)
- Expiry alerts for driving licences and crew training (BLS/ACLS)
- Crew check-in and vehicle checklist in the app at shift start

Booking and dispatch

- Booking channels: emergency helpline, front desk, doctors (inter-hospital transfers), patient app, website
- Trip types: emergency pickup, hospital-to-home drop, inter-hospital transfer, recurring trips (dialysis, chemotherapy), event standby, hearse
- Pickup location by address, map pin, or a location link sent to the caller
- **Smart dispatch:** suggests the nearest available ambulance of the right type
- Crew receive an app alert with navigation as soon as a trip is assigned
- Advance booking of scheduled trips (e.g. regular dialysis patients)

Live tracking

- Live location from the crew app or a GPS device
- Control-room map showing every ambulance and its status
- Tracking link and ETA sent to the family on WhatsApp
- Trip stages with timestamps: assigned → departed → at pickup → patient on board → at hospital → completed

Link to emergency

- Crew enter patient details, vitals and condition (triage) in the app en route
- The emergency team is alerted before arrival so the bed and team are ready
- Quick registration before arrival (UHID created)
- MLC flag and required details for accident cases

Billing

- Rate card by type: base fare, per km, waiting charge, night charge, equipment/oxygen charge
- Trip distance taken from GPS automatically

- Charges added to the patient's hospital bill or billed separately
- TPA/insurance and government scheme cases (e.g. 108 tie-ups)
- Payout calculation for partner/vendor ambulances

Maintenance and costs

- Fuel entries (litres, amount, odometer) and mileage reports
- Service schedules and reminders (by km or days), breakdown complaints
- Oxygen refills and drug stock, linked to pharmacy/store
- Per-vehicle costs posted to accounts (module 16)

Reports

- Response time (call to pickup), trips per ambulance, distance covered, revenue versus cost, crew-wise trips, idle time

18. CRM and call centre

A complete CRM for the whole patient lifecycle, not just new leads, with a built-in cloud call centre. Every call, message, visit and bill lands on one patient timeline, so staff always know who is calling and what they need. Module 13 captures leads; this module manages every relationship after that.

Cloud telephony and IVR

- One or more hospital numbers (toll-free or local) through a cloud telephony connector (Exotel, Knowlarity, Ozonetel, MyOperator); providers can be switched
- Configurable IVR menus (e.g. "press 1 for appointments, 2 for reports, 3 for emergency"), business hours and holiday messages
- Call routing by department, language, skill or round-robin; emergency calls go straight to the emergency desk and ambulance control room (module 17)
- Missed calls trigger an automatic callback task and a WhatsApp message
- Voicemail after hours, transcribed into a task

Agent console

- **Screen pop:** an incoming call opens the caller's record automatically (UHID, recent visits, upcoming appointments, dues, open complaints); unknown numbers open a new lead form
- Book, reschedule or cancel appointments, share report links and send payment links during the call
- Call disposition codes (booked, enquiry, complaint, follow-up, wrong number) and mandatory call notes
- Warm transfer and conference to a doctor, department or supervisor

- Agent status: available, on call, on break, after-call work

Outbound calling

- Calling lists: follow-ups due, missed appointments, pending reports, unpaid bills, lead callbacks
- Progressive and preview dialer that dials the next number automatically
- Do-not-call list and calling-hour rules to stay compliant
- Every outbound call logged against the patient or lead

Call records and quality

- Call recording with consent announcement, stored securely with retention rules
- Recordings playable from the patient timeline (role-restricted)
- Supervisor live listen, whisper and barge-in
- Quality scoring of calls against a checklist

Patient 360 and segments

- One timeline per patient: visits, bills, prescriptions, calls, WhatsApp/SMS/email, feedback, complaints, campaigns
- Tags and segments built from data, e.g. "diabetic patients", "no visit in 6 months", "senior citizens in Noida", "insured patients"
- Segments refresh automatically as data changes

Campaigns and automation

- Broadcast campaigns on WhatsApp, SMS and email to any segment, with templates and scheduling
- Automated journeys: vaccination due, annual health check-up, chronic disease follow-up, medicine refill reminders, post-discharge check-in calls on day 1, 7 and 30
- Birthday and festival greetings
- Opt-out handling and consent tracking (DPDP Act)
- Campaign results: delivered, read, clicked, booked, revenue

Memberships, loyalty and referrals

- Health cards and family membership plans with discounts and validity
- Loyalty points earned on bills and redeemed at billing
- Patient referral programme with referral codes and rewards

Referring doctor network (B2B)

- Directory of external referring doctors and clinics with specialty, location and potential

- Referral tracking: patients referred, revenue generated, trend by month
- Field team (marketing executives): visit planning, check-in with GPS, visit notes, targets
- Referral payouts linked to doctor payouts (module 8), within the law and NMC rules

Corporate and partner accounts

- Corporate clients, insurance and TPA relationships, camps and tie-ups with contacts, contracts and renewal dates
- Activity log and follow-up tasks per account

Complaints and feedback

- Feedback after OPD visits and discharge by WhatsApp/SMS, including NPS
- Low scores automatically open a complaint ticket
- Ticketing with categories, priority, owner, SLA timers and escalation to management
- Root-cause and resolution reports (feeds NABH quality indicators, module 14)

Call centre and CRM reports

- Calls received, answered, missed, abandoned; average wait and handling time; peak hours
- Agent performance: calls handled, bookings made, quality score
- Conversion: enquiry to appointment to visit to revenue
- Campaign ROI, referral doctor contribution, complaint resolution time, NPS trend

New roles: Call centre agent, Call centre supervisor, CRM/marketing manager, Field executive.

19. Inventory and supply chain

One inventory system for everything the hospital buys, stores and uses: medicines, consumables, implants, reagents, gases, linen, food, stationery and IT items. A central store supplies sub-stores across the hospital, every movement is recorded, and stock for any item is visible in every store at any moment. Pharmacy dispensing (module 5) and OT/ICU consumption draw from this module.

Item master and categories

- Categories: medicines, surgical consumables, implants, lab reagents and kits, radiology films and contrast, medical gases, OT items, CSSD instruments, housekeeping, linen, kitchen/food, stationery, IT and electricals, furniture
- Per item: code, name, generic/brand, category, unit of purchase vs unit of issue (e.g. box of 100 → 1 piece), HSN, GST rate, storage condition, batch/expiry tracked or not, serial-number tracked or not, patient-chargeable or not

- Item classification for control: ABC (value), VED (vital/essential/desirable), FSN (fast/slow/non-moving)
- Barcode/QR on items and shelves

Hospital's own serial numbers and labels

- Every item can carry a hospital-generated unique ID, printed as a barcode/QR label at goods receipt, in addition to the manufacturer's serial or batch number
- Configurable number format per category, e.g. `NOIDA/AST/2026/000123` for assets or `CS/IMP/000045` for implants (branch, category, year, running number)
- Tracking level chosen per item: **unit-level** (every piece has its own ID, e.g. assets, implants, equipment, instruments), **batch-level** (one label per batch, e.g. medicines, syringes, gloves), or **none** (low-value items)
- Label printing on standard barcode printers, with reprint and damaged-label replacement logged
- Scanning the label anywhere shows the item's full history: GRN, vendor, every store it moved through, who it was issued to, which patient it was used on, maintenance and current location
- Duplicate IDs are blocked; IDs are never reused, even after disposal

Store structure

- Multiple store types: central store, pharmacy (OPD/IPD), OT store, ICU store, ward sub-stores, lab store, radiology store, CSSD, kitchen store, housekeeping store, general/admin store
- Store hierarchy per branch, and inter-branch transfers
- Bin/rack/shelf locations inside each store
- Separate stock, min/max levels and reorder points per store

Procure-to-consume flow

1. **Indent:** a department raises a request for items (manual, or auto-suggested from min levels)
2. **Approval:** by department head and store, with value-based limits
3. **Purchase requisition and quotation:** compare vendor quotes, or use a rate contract
4. **Purchase order:** sent to the vendor by email/WhatsApp, with delivery date and terms
5. **Goods receipt (GRN):** quantity, batch, expiry, MRP and free quantity checked against the PO; quality check; rejection and short supply recorded
6. **Put-away:** stock lands in the right store and bin
7. **Issue:** central store to sub-store, or store to department/patient
8. **Consumption:** usage recorded against department, procedure or patient

9. **Returns:** department to store, store to vendor (expired, damaged, recalled)

Medical stock controls

- Batch and expiry tracking with **FEFO** (first-expiry-first-out) suggested at issue
- Near-expiry alerts (e.g. 30/60/90 days) and return-to-vendor before expiry
- Narcotic and psychotropic drugs (Schedule H1/X, NDPS): separate register, double sign-off at issue, balance checks
- Cold chain: temperature logs for vaccines and refrigerated items, with alerts on out-of-range readings
- Batch recall: find every store and patient that received a recalled batch
- Medical gases: cylinder tracking (full, in use, empty), manifold and refill records

Patient-linked consumption

- OT, ICU, ward and emergency consumables charged to the patient's bill automatically when issued
- Implants tracked by serial number and batch, linked to the patient record and surgery (for recalls and warranty)
- Procedure kits/packages (e.g. "normal delivery kit") issued as a set with standard consumption
- Variance between standard and actual consumption per procedure

Vendors and contracts

- Vendor master with drug licence, GSTIN, rating and lead time
- Rate contracts with validity; PO prices checked against the contract
- **Consignment stock:** implants and high-value items held at the hospital and purchased only when used
- Vendor performance: on-time delivery, rejection rate, price trend
- Vendor bills matched to PO and GRN (3-way match) before payment in accounts (module 16)

Stock audit and valuation

- Cycle counts and full physical stock-takes using barcode scanners or the mobile app
- Variance report (system vs physical) with approval for adjustments
- Wastage, breakage and damage records with reasons
- Stock valuation by FIFO or weighted average; all movements post to accounts automatically

Non-medical and admin inventory

- Stationery, printing, housekeeping chemicals, uniforms, IT consumables, kitchen supplies on the same indent-issue flow
- Department-wise budgets and consumption limits

Asset management

- Asset register for equipment, furniture, IT and vehicles: tag, location, custodian, purchase value, warranty
- AMC/CMC contracts, preventive maintenance schedule, breakdown tickets, downtime tracking (links to Biomedical in module 14)
- Asset transfers between departments and branches
- Depreciation schedule posted to accounts; asset disposal and write-off

Inventory reports

- Current stock by store, item and batch; stock value
- Consumption by department, procedure and patient; cost per patient-day
- Reorder list, near-expiry list, expired-stock loss
- Slow-moving and dead stock, ABC/VED analysis
- Purchase register, vendor-wise purchases, price variance

New roles: Store manager, Purchase officer, Sub-store in-charge (OT/ICU/ward), Biomedical engineer.

20. Incidents, escalation and medico-legal

When a case goes wrong, a family complains, or a legal notice arrives, the hospital needs a fast, documented and confidential response. This module handles clinical incidents, escalations, medico-legal cases and every kind of dispute, and protects the records that will matter later. Access is restricted to named people (medical superintendent, quality, legal, management).

Incident and adverse event reporting

- Any staff member can report an incident in under a minute from web or app, with an option to report anonymously
- Incident types: medication error, patient fall, hospital-acquired infection, surgical complication, wrong-site/wrong-patient event, transfusion reaction, unexpected death, equipment failure, violence against staff, needle-stick injury, others (configurable)
- Severity levels: near miss, no harm, mild, moderate, severe, death (sentinel event)
- Link the incident to the patient, staff, department, equipment and batch involved
- Attach photos, documents and statements

Escalation matrix

- Configurable escalation ladder by incident type and severity: who is alerted, at which level, and within how many hours (e.g. department head → medical superintendent → CEO/owner → board)
- Automatic alerts by app, WhatsApp and SMS; unacknowledged alerts escalate to the next level
- Sentinel events alert top management immediately
- Escalation timeline recorded with timestamps for every step

Investigation and root cause analysis (RCA)

- Assign an investigation owner and team, with deadlines
- Structured RCA: timeline of events, contributing factors, root causes (5 Whys / fishbone)
- Corrective and preventive actions (CAPA) with owner, due date and verification of completion
- Mortality and morbidity (M&M) review meetings: case list, minutes, decisions
- Learnings shared across departments without naming individuals

Family communication and disclosure

- Log every conversation with the patient or family: who spoke, what was explained, who was present
- Counselling notes, high-risk consent records and second-opinion requests linked to the case
- Templates for formal communication letters

Medico-legal case management

- Register of legal matters: legal notices, consumer court cases, civil and criminal cases, police complaints, medical council complaints, regulator queries
- Case file: case number, forum/court, parties, claim amount, advocate, internal owner, status, next hearing date
- Hearing calendar with reminders, and outcomes recorded per hearing
- Legal expenses, settlements and compensation tracked and posted to accounts (module 16)
- Link to MLC records from emergency (module 14) and police intimations
- Professional indemnity insurance: policies for the hospital and doctors, intimation to the insurer, claim status

Legal hold and evidence preservation

- Place a **legal hold** on any patient or case: records are locked against editing and deletion, beyond normal retention periods
- Every version of every record is preserved with the audit trail (who changed what, when)
- One-click case bundle for advocates and courts: registration, consent forms, clinical notes, nursing charts, orders, lab and radiology reports, bills, call recordings and communication logs, in chronological order, as a paginated PDF
- References to CCTV footage and physical evidence, with custody records
- Certified copies of medical records issued with request logs (who asked, what was given, when)

Dispute management

Dispute type	Flow
Patient billing dispute	Patient raises a query → bill review by billing and the treating doctor → justification, correction, discount or refund with approval → response to patient
Insurance / TPA dispute	Claim deduction or rejection → internal review → appeal to TPA/insurer with documents → grievance to insurer/ombudsman if needed → outcome posted to the claim (module 7)
Doctor payout dispute	Doctor raises a query on the statement → finance review → adjustment or explanation (module 8)
Vendor dispute	Short, wrong or defective supply, or payment disagreement → debit note, replacement or settlement (module 19)
Staff grievance	Employee complaint → HR review → committee if needed → resolution (module 15)
POSH case	Confidential complaint to the Internal Committee → inquiry → findings and action, restricted to committee members, with statutory timelines

Every dispute has an owner, SLA, status, documents and a full history.

Reports

- Incidents by type, severity, department and month; near-miss reporting rate
- Open RCAs and overdue corrective actions

- Open legal cases, upcoming hearings, legal spend and exposure
- Disputes by type, resolution time and outcome
- NABH and quality indicators fed automatically (module 14)

New roles: Quality manager, Medico-legal officer, Legal counsel (view and case files), Internal Committee member (POSH cases only).

21. Clinical safety and nursing

Most serious hospital errors happen around medication and handovers. This module builds safety checks into the doctor's and nurse's daily work, and gives nursing a complete digital workflow. NABH assessors look for exactly these controls.

Computerized order entry (CPOE)

- All doctor orders entered digitally: medicines, IV fluids, lab, radiology, diet, nursing instructions, procedures, consults
- Order sets for common conditions and protocols (e.g. chest pain, sepsis, post-op care)
- Verbal/telephone orders recorded by the nurse and countersigned by the doctor within a set time
- Every order shows status: ordered → verified → in progress → done

Clinical decision support

- Drug–drug interaction, drug–allergy and duplicate-therapy alerts while prescribing
- Dose checks by age, weight, kidney/liver function, with paediatric weight-based dosing
- High-alert medication flags (insulin, heparin, potassium chloride, chemotherapy) with double-check requirements
- Pregnancy and lactation warnings
- Critical lab value alerts to the treating doctor, with acknowledgement tracking
- Alert overrides require a reason and are reported

Medication administration (eMAR)

- Pharmacist verification of IPD orders before dispensing
- Bedside barcode scanning of the patient wristband and the medicine (five rights: patient, drug, dose, route, time)
- Due, given, held, refused and missed doses recorded with reason; overdue-dose alerts
- PRN (as needed) doses with effect noted
- Wristbands with barcode/QR printed at admission

Early warning and deterioration

- Automatic early warning score (NEWS2; PEWS for children; MEOWS for obstetrics) from vitals
- Escalation to the doctor and rapid response team when scores cross thresholds
- Code blue activation and resuscitation record
- Sepsis screening prompts

Nursing workflow

- Nurse–patient assignment per shift and ward census
- Admission nursing assessment
- Risk assessments with scheduled reassessment: fall risk (Morse), pressure ulcer (Braden), pain scale, nutrition (MUST), DVT risk
- Nursing care plans: problem, goal, interventions, evaluation
- Shift handover in SBAR format, per patient
- Intake/output, wound care, catheter and line care (insertion date, due for change)
- Restraint and seclusion records with monitoring intervals
- Nursing tasks list per shift with completion tracking

Patient safety checks

- WHO surgical safety checklist (sign in, time out, sign out) mandatory before closing an OT record
- Patient identification with two identifiers on every order and sample
- Transfusion checklist with bedside verification

Reports

- Medication errors and near misses, alert override rates, missed doses
- Falls, pressure ulcers, early warning escalations, code blue outcomes
- Nursing workload and staffing ratio by ward

22. Infection control, biomedical waste and statutory compliance

Several of these are legal requirements in India, and lapses can lead to penalties, sealing of equipment or loss of licence. The module keeps the hospital compliant by recording what the law requires at the point of work and warning before anything lapses.

Infection control (HICC)

- Surveillance of hospital-acquired infections: CAUTI, CLABSI, VAP, surgical site infection, with device-days tracked automatically from IPD records
- Infection rates per 1,000 device-days by ward and ICU

- Culture and sensitivity results flagged for multi-drug-resistant organisms; isolation alerts on the bed board
- Hand hygiene audits, environmental swab results, needle-stick and exposure records
- Outbreak alerts when cases cluster in a ward
- HICC committee meetings, minutes and actions

Antibiotic stewardship

- Restricted antibiotics require justification or ID-specialist approval
- Automatic stop/review dates on antibiotic orders
- Antibiogram (local resistance patterns) generated from lab data
- Antibiotic usage reports (DDD per 100 patient-days) by department

Biomedical waste (BMW Rules, 2016)

- Colour-coded waste categories (yellow, red, white, blue) per ward
- Bag-level barcode labels with ward, category and weight at generation
- Handover to the authorised common BMW treatment facility (CBWTF) with weights, vehicle and signature
- Daily waste register, monthly records and the annual report to the State Pollution Control Board
- Mismatch alerts between generated and handed-over weights
- Staff training and immunisation records for waste handlers

Clinical statutory registers

- **PCPNDT Act:** Form F completed for every ultrasound on a pregnant woman, before the scan report can be signed; registers, monthly reports to the appropriate authority; machine registration records
- **MTP Act:** MTP register, opinion forms and consent, with confidentiality controls
- **AERB:** radiology equipment licences, radiation worker dosimetry (TLD) records, QA test schedules
- **NDPS:** narcotic registers (from module 19)
- **Notifiable diseases:** reporting to IDSP/IHIP for notifiable conditions (e.g. dengue, malaria, TB under Ni-kshay)
- Births and deaths (module 14) and MLC records (module 20) linked here

Licence and compliance calendar

- Register of every licence and certificate with issuing authority, number, validity and owner: Clinical Establishment registration, fire NOC, BMW authorisation, pharmacy/drug licences, blood bank licence, PCPNDT registration, AERB licences, lift and boiler certificates, pollution consents, NABH/NABL accreditation, trade licence

- Renewal alerts at 90/60/30/7 days, escalating to management
- Document storage of current and past certificates
- Inspection and audit records with findings and closure

Reports

- Infection rates, MDRO cases, antibiotic usage, hand-hygiene compliance
- Waste generated per bed per day, by category
- Compliance status board: valid, expiring, expired

23. Hospital KPIs, MIS and analytics

Management runs the hospital on a small set of numbers. This module computes them automatically from operational data, so nobody prepares reports by hand, and lets owners drill from a group total down to a single bill.

Standard hospital KPIs

Area	KPIs
Beds	Bed occupancy %, average length of stay (ALOS), bed turnover rate, discharges before noon
Revenue	ARPOB (average revenue per occupied bed), revenue per OPD patient, payer mix (cash/TPA/scheme/corporate), revenue by department and doctor
OPD	Footfall, new vs follow-up, average waiting time, consultation time, no-show rate
OT	OT utilisation %, surgeries per OT per day, first-case on-time start, cancellations
Emergency	Door-to-doctor time, admission rate, left without being seen
Diagnostics	Test volumes, turnaround time (TAT) vs target, repeat rate
Quality	Readmissions within 30 days, mortality rate, infection rates, incidents, patient satisfaction (NPS)
Finance	Collections, receivables days, TPA outstanding, cost per patient-day, EBITDA by department

Area	KPIs
People	Staff per occupied bed, nurse-to-patient ratio, overtime cost, attrition

Dashboards

- Role-based dashboards: owner/CEO, medical superintendent, department head, finance, nursing, operations
- Drill-down: group → branch → department → doctor → patient → bill
- Daily summary on WhatsApp/email to owners each morning
- Targets and budgets per KPI with variance highlighting
- Trend charts by day, month and year, and branch comparison

Analytics tools

- Report builder: pick fields, filters and grouping without code; save and schedule reports
- Export to Excel/PDF; scheduled email delivery
- Department profitability: revenue minus direct costs (consumables, staff, doctor share) and allocated overheads
- Read-only data access for external BI tools (Power BI, Metabase) for large customers

24. Health check-ups, home care and pharmacy delivery

These services bring in extra revenue and keep patients loyal. Each runs on the same patient record, billing and inventory as the rest of the platform.

Health check-up workflow

- Package booking online, by phone or at the front desk, with pre-test instructions (fasting) sent on WhatsApp
- On arrival, a single registration creates all orders in the package
- **Smart routing:** the system sends the patient to the next free station (blood collection, ECG, X-ray, USG, eye, dental, doctor) to cut waiting
- Live tracker of every check-up patient and pending stations
- Consolidated report: all results in one PDF with the doctor's summary and recommendations
- Corporate check-ups: employee lists, bulk scheduling, company billing, aggregated (anonymised) health report for the employer
- Follow-up reminders for abnormal results

Home care services

- Services: home sample collection, home nursing, home physiotherapy, doctor home visits, home ICU/ventilator setup, post-discharge care, elderly care
- Booking by app, website, phone or WhatsApp with slot and address
- Field staff assignment by location and skill, with route view
- Field app: visit check-in with GPS, vitals and notes, sample barcode scanning, photos, patient signature
- Sample chain of custody from home to lab, with temperature where required
- Collection of payment at home (UPI/card/link)
- Recurring visit plans (e.g. daily nursing for 14 days) and package billing

Pharmacy home delivery

- Order medicines from the patient app, website or WhatsApp; upload a prescription or reorder from a past one
- Pharmacist verification of the prescription before dispatch (Schedule H/H1/X rules enforced)
- Delivery assignment, tracking link, proof of delivery
- Payment online or cash/UPI on delivery
- Refill reminders for chronic medicines
- Stock deducted from the pharmacy store (module 19)

25. Radiology information system (RIS)

Imaging is a high-revenue, high-volume department with expensive machines. The RIS schedules every scan, feeds the machines their worklists, and gives radiologists a reporting queue, with images viewable from anywhere through PACS.

- **Scheduling:** modality-wise calendars (X-ray, USG, CT, MRI, mammography, DEXA, fluoroscopy, cath lab), slot lengths per study, machine downtime blocks
- **Preparation:** patient instructions (fasting, contrast, full bladder) sent on WhatsApp; contrast allergy and kidney function (creatinine) checks before contrast studies; pregnancy check for women of child-bearing age
- **Modality worklist (DICOM MWL):** patient details sent to the machine, so technicians never retype names or IDs
- **Study tracking:** scheduled → arrived → in progress → images acquired → reported → verified → delivered
- **Radiologist reporting queue:** prioritised by urgency (emergency, IPD, OPD); structured templates per study; voice dictation; comparison with prior studies; critical-finding alerts to the referring doctor

- **Teleradiology:** remote radiologists report from anywhere, with per-study payout; studies not available in-house (e.g. PET-CT) go to partner centres under the same tie-up, rate, tracking and reconciliation model as partner labs (module 4)
- **PACS integration:** web image viewer link from the patient record, the report and the patient app
- **Consumables:** contrast, films and syringes deducted from the radiology store (module 19)
- **Compliance:** PCPNDT Form F for obstetric ultrasound and AERB records (module 22)
- **Reports:** studies per modality, machine utilisation, TAT from scan to report, radiologist productivity, repeat/reject rate

26. Facility services

The non-clinical services that keep a hospital running, each with requests, tracking and costs in the same system.

Facility maintenance

- Complaint tickets for electrical, plumbing, HVAC/AC, carpentry, civil, IT, lifts, medical gas pipeline
- Raised from web, app or QR codes on room doors; photo attachment
- Assignment to technician or vendor, priority, SLA and escalation
- Preventive maintenance schedules for generators, chillers, lifts, fire systems, water tanks
- Utility readings (electricity, water, diesel) and consumption trends
- Spare parts drawn from the store (module 19); costs posted to accounts

Mortuary management

- Body receipt with identification tags, cause of death reference, MLC flag
- Cold chamber allocation and temperature log
- Post-mortem request and police documentation for MLC cases
- Release to family with ID verification, documents checklist and signature
- Hearse booking (module 17)

Canteen and cafeteria

- Menu and pricing, point-of-sale billing for visitors and staff
- Staff meal deductions through payroll (module 15)
- Patient diet orders served from the same kitchen (module 14)
- Kitchen stock from the store (module 19)

Parking

- Entry/exit tickets or FASTag/number-plate integration, hourly rates, staff and doctor passes
- Validation discounts for patients

Document management (DMS)

- Scan old paper records, outside reports and ID documents into the patient file, indexed by type and date
- OCR so scanned text is searchable
- Retention rules per document type, with legal hold support (module 20)
- Bulk scanning for migrating historical records at go-live

Product principle: configuration, not code

Any customer (dental clinic, eye clinic, multi-specialty clinic or 500-bed hospital) must be able to set the product up **on its own**, without Finmeyo writing new code. If a new customer needs code changes, the product has become a custom project. Everything that varies between customers is therefore a setting.

What varies between customers	How the customer configures it	Example
Departments and specialties	Department master	One has dental and ENT, another has 12 departments
Services and rates	Service master and rate lists	Separate rates for general, corporate, insurance, senior citizens
EMR forms	No-code form builder	Refraction form for an eye clinic, cycle tracking for an IVF centre
Prescription and report layout	Print template builder	Letterhead, logo, doctor's signature, A4/A5/thermal
Roles and permissions	Role builder (checkbox matrix)	In a small clinic the receptionist is also the cashier
Workflow	Workflow settings	Pay before consultation, or after
Bill and UHID numbering	Numbering series settings	`BLR/OPD/2026/00045`

What varies between customers	How the customer configures it	Example
Tax	Tax engine (rate, exempt/taxable, HSN/SAC)	Admin updates GST rates when they change
Messages	Notification templates (WhatsApp/SMS/email)	Reminders in the clinic's own words
Modules	On/off per plan	Small clinic: IPD off; hospital: everything on
Extra information	Custom fields on any form	"Referred by", "Company ID" on the patient record
Language	Language setting (per user)	English at launch; later front desk in Hindi, doctors in English

Rule for the team: before building any feature, ask "is this the same for every customer?" If not, make it a setting; never hard-code it.

Masters (setup data)

Masters are the data a customer enters once and the whole platform runs on. Every master supports Excel import and export. Some masters (medicines, lab tests, ICD-10) come pre-filled so new customers can go live quickly.

Master	Key fields	Pre-filled?
Organization	Name, logo, GSTIN, PAN, Clinical Establishment registration no., address, contact	No
Branch	Name, code, address, GSTIN (if separate), timings, letterhead	No
Department	Name, code, type (clinical/lab/pharmacy/admin), head of department	Yes, common list
Doctor	Name, qualifications, registration no. (NMC/State Council), specialty, department, signature image, consultation fee, schedule, payout rule	No
Staff / user	Name, role, branch, mobile, email, login status	No
Service	Name, code, department, rate, SAC code, GST type (exempt/taxable), doctor share %	Yes, common list

Master	Key fields	Pre-filled?
Rate list	Name (general/corporate/TPA/staff), service-wise rates, valid from–to	No
Package	Name, included services, package rate, validity	Some samples
Lab test	Name, code, sample type, container, parameters, unit, normal ranges (age/gender), method, TAT, rate	Yes
Radiology test	Name, modality (X-ray/USG/CT/MRI), report template, rate	Yes
Medicine / item	Brand, salt (generic), strength, form, manufacturer, HSN, GST rate, schedule (H/H1/X), pack size	Yes, large catalogue
Supplier	Name, GSTIN, drug licence no., contact, payment terms	No
Ward / room / bed	Ward, room, bed no., type, daily charge, nursing charge	No
Insurance / TPA	Company, TPA, contact, rate list, claim format	Yes, common list
Corporate	Company, employee rate list, credit limit, billing cycle	No
Referring doctor	Name, contact, commission rule	No
Diagnosis	ICD-10 code and name, clinic favourites	Yes
Payment modes	Cash, UPI, card, cheque, wallet, insurance, credit	Yes
Tax	Tax name, rate, valid from, HSN/SAC mapping	Yes (GST)

Form and template builder (no code)

This feature is what makes the product sellable to every specialty. Finmeyo ships ready templates for all major specialties, and customers can modify them or build their own.

EMR form builder

- Drag-and-drop forms made of sections and fields
- **Field types:** text, number (with unit), date, dropdown, multi-select, checkbox, yes/no, scale (1–10), table/grid, image upload, marking on a body diagram, dental chart, calculated fields (e.g. BMI)

- Field rules: required/optional, normal ranges, conditional display ("if X = yes, show Y")
- Carry forward values from the previous visit (e.g. weight, current medicines)
- Form versioning: old visits keep showing in their original version
- Forms linked to department, doctor or visit type
- Template library: forms can be copied across branches

Specialty coverage

Ready templates for all the specialties below ship in Release 1.0. Each is built with the form builder, so hospitals can change them and add new specialties without code.

Specialty	Specialty-specific content
General medicine	Chief complaint, history, examination, diagnosis, chronic disease tracking (BP, sugar)
Paediatrics	WHO growth charts, vaccination schedule and reminders, weight-based dosing
Obstetrics and gynaecology	ANC card, EDD calculator, USG records, labour/partograph, delivery record, newborn record
IVF / fertility	Cycle tracking, follicle monitoring, embryology lab record, consent forms
General surgery	Pre-op assessment, operation notes, post-op care, wound record
Orthopaedics	Joint examination, fracture record, implant details (batch/serial), physiotherapy plan
Cardiology	ECG/echo/TMT reports, cath lab procedure notes, stent details, risk scores
Neurology / neurosurgery	Neurological examination, GCS, stroke scale, EEG/EMG reports
Nephrology and dialysis	Dialysis sessions, machine allocation, pre/post weight, lab trends
Oncology	Cancer staging, chemotherapy protocols and cycles, day-care chemo, tumour board notes
Gastroenterology	Endoscopy/colonoscopy reports and images, liver function trends

Specialty	Specialty-specific content
Pulmonology	Spirometry/PFT, sleep studies, oxygen therapy records
Urology	Uroflowmetry, stone records, prostate scores
ENT	Ear/nose/throat examination diagrams, audiometry
Ophthalmology	Vision, refraction, IOP, slit lamp, fundus images, spectacle prescription, cataract surgery record
Dental	FDI tooth chart, treatment plan, multi-sitting treatments, lab work (crowns) tracking
Dermatology and cosmetology	Lesion marking on body diagrams, before/after photos, session-based procedures
Psychiatry	Mental status examination, assessment scales (PHQ-9, GAD-7), confidential notes
Physiotherapy and rehabilitation	Assessment, session plan, exercise chart, progress tracking
Endocrinology / diabetology	Sugar/HbA1c trends, insulin chart, diet plan
Emergency medicine	Colour-coded triage, quick registration, MLC record
Critical care (ICU)	Hourly ICU chart, ventilator settings, APACHE/SOFA scores, infusion record
Anaesthesia	Pre-anaesthesia check-up, anaesthesia record, recovery notes
Radiology	Report templates (X-ray, USG, CT, MRI), PACS viewer link
Pathology	Test parameters, normal ranges, histopathology reports
Dietetics	Diet plans, ward diet orders

Print and document template builder

- Prescriptions, bills, receipts, lab reports, discharge summaries, consent forms, certificates (fitness, medical leave)
- Letterhead, logo, footer, doctor's signature and stamp, QR code for report verification
- Paper sizes: A4, A5, Letter, 80 mm thermal

- Placeholders such as `{patient_name}`, `{uhid}`, `{doctor_name}`, with preview
- Separate templates per branch or doctor

Custom fields

- Add fields to any record: patient, appointment, bill, admission
- Custom fields are available in reports and search

Doctor productivity shortcuts

- Favourite prescription and advice sets (e.g. "Viral fever set")
- Shorthand: typing `1-0-1` expands to "morning and night"
- Voice dictation for notes

Workflow, numbering, approvals and notifications

Every hospital works differently. All of this is decided by settings, not code.

Workflow settings

- OPD: **pay before seeing the doctor**, or **see the doctor, then pay**
- Who records vitals: nurse, front desk, or the doctor
- Lab: whether payment is required before sample collection
- Pharmacy: prescription-only dispensing, or counter sales too
- IPD: whether a deposit is required at admission, and the minimum amount
- Reports reach the patient only after doctor/pathologist approval
- Appointment slot type: fixed time, token, or both

Numbering series

- Separate formats for UHID, bills, receipts, IPD numbers, lab samples and invoices
- Formats can include branch code, year, financial year, department and a running number
- Optional reset every financial year (1 April)
- GST invoice numbers without gaps or duplicates

Approvals

- Discounts: front desk up to a limit, manager approval above it
- Bill cancellations and refunds only after approval, with a mandatory reason

- Purchase orders approved by amount
- Approval requests delivered in the app and on WhatsApp

Notification rules

Event	Recipient	Channel (customer chooses)
Appointment booked/cancelled	Patient, doctor	WhatsApp, SMS, app
Appointment reminder (X hours before)	Patient	WhatsApp, SMS
Report ready	Patient, doctor	WhatsApp, app
Bill / payment receipt	Patient	WhatsApp, email
Follow-up due	Patient	WhatsApp
Critical lab value	Doctor	App push, SMS
Low stock / expiry	Pharmacist, admin	App, email
Discount/refund approval	Manager	App, WhatsApp
Daily collection summary	Owner	WhatsApp, email

Customers can edit every message template and switch each event on or off.

Integrations and open API

Each integration is built as a connector. Customers enable it in settings with their own API keys and can switch providers when needed, so the platform is never locked to one vendor.

Integration	Purpose	Provider
Payment gateway	Online payments, UPI QR, payment links	Razorpay at launch; others can be added later
Card machine / UPI soundbox	Counter payments auto-matched to bills	Pine Labs, bank POS

Integration	Purpose	Provider
WhatsApp Business API	Reminders, reports, bills, booking	Any approved provider (Gupshup, Interakt, Meta direct)
SMS	OTP and alerts (DLT registered)	MSG91, Twilio, Kaleyra
Email	Bills, reports, monthly statements	SES, SendGrid, SMTP
Insurance claims	Pre-auth and claim submission, status	NHCX; ready packets for TPA portals
ABDM	ABHA ID, facility/doctor registries, health record sharing	ABDM gateway (M1/M2/M3)
Lab analyzers	Results directly into the system	HL7 / ASTM interface
PACS / imaging	View X-ray, CT, MRI images	DICOM viewer, PACS link
Accounting export	Sync for the CA	Tally (XML), Zoho Books
e-Invoice / GST	IRN generation (larger hospitals)	GSP provider
Biometric attendance	Staff attendance	ZKTeco-type devices
Video calls	Teleconsultation	Jitsi, Zoom, Agora
Maps	Branch locations, patient addresses, ambulance tracking	Google Maps API
Ad platforms	Lead ads and conversion tracking	Meta, Google Ads
Cloud telephony	IVR, call routing, screen pop, recording, dialer	Exotel, Knowlarity, Ozonetel, MyOperator

All integrations ship in Release 1.0.

Open API and webhooks

- A documented **REST API** for every module, so large hospitals and partners can connect their systems
- **Webhooks** for events such as "appointment created", "bill generated", "report ready"
- Per-customer API keys, rate limits and access logs
- Benefit: the answer to "will it connect to our system?" is always yes

Language, tax and international readiness

The launch market is India, but the platform is built from day one so it can later be sold in Nepal, Bangladesh, the Middle East or Africa. Designing for this now is cheap; retrofitting it later is expensive.

- **Language:** English only at launch. All interface text lives in resource files (not in code), so Hindi and regional languages can be added later by translation alone, with each user choosing their language.
- **Patient documents:** in a later release, prescriptions and advice can print in the patient's language
- **Tax engine:** GST is not hard-coded. Tax names, rates, exempt/taxable rules and invoice formats are settings, so VAT and other regimes can be supported.
- **Currency:** each organization has its own currency (₹, NPR, AED); amounts can display in lakh/crore or million format
- **Date, time and time zone:** configurable date format; each branch has its own time zone
- **Phone numbers and addresses:** country codes on phone numbers; address formats per country
- **Medical coding:** ICD-10 by default, with an ICD-11 option later
- **Data residency:** option to keep each country's data on servers in or near that country

Platforms

The web app runs every module. The mobile apps, displays and WhatsApp are focused front ends on the same system, not separate products.

Platform	Users	Capabilities	Release
Web app (browser)	All hospital staff and admins	Every module; works on desktop and tablet	1.0
Doctor mobile app	Doctors	Today's appointments, patient history, prescriptions, lab reports, IPD round notes, critical alerts, own earnings	1.0
Nurse tablet view	Ward and ICU nurses	Vitals, medication chart, nursing notes, bedside entry	1.0
Staff self-service app	All employees	Attendance, roster, leave, payslips	1.0

Platform	Users	Capabilities	Release
Ambulance crew app	Drivers and paramedics	Trip alerts, navigation, live location, en-route patient details	1.0
WhatsApp	Patients and staff	Booking, confirmations, reminders, report and bill PDFs, follow-ups, feedback	1.0
TV / kiosk	Waiting areas	Token/queue display, self check-in kiosk	1.0
Patient app	Patients	Appointments, online payment, reports, prescriptions, bills, family members, teleconsultation	1.0
Field staff app	Home care staff, phlebotomists, delivery staff, field executives	Visit/delivery list, route, GPS check-in, sample scanning, vitals, photos, signatures, payment collection	1.0

Technical notes

- The web app is responsive, so front desks can run on tablets
- Mobile apps for Android and iPhone from a single codebase (React Native or Flutter)
- WhatsApp through the official WhatsApp Business API via an approved provider; each message has a small cost that must be built into pricing
- A basic offline mode so the front desk keeps working during internet outages

Accessibility: the patient app, websites and kiosks follow WCAG 2.1 AA: readable font sizes, high contrast, screen-reader support and simple flows for elderly and low-vision users.

SaaS and subscription

One platform serves every customer, but each customer's data is fully isolated (multi-tenancy).

Pricing: per clinic now, configurable later

At launch each hospital/clinic (per branch) pays a fixed monthly or yearly fee. The model may change once the market responds, so pricing lives in a **pricing engine**, not in code, and the super-admin can create new plans without development.

Pricing model	Basis	Status
Per clinic / per branch	Fixed fee per branch	Used at launch
Per login (user)	Number of active users	Built into the engine, enabled later
Per doctor	Number of active doctors	Built into the engine
Per bed	Number of hospital beds	Built into the engine
Module add-ons	Separate fee for selected modules (e.g. blood bank, website)	Built into the engine
Usage-based	WhatsApp/SMS messages, storage, claim volume	Built into the engine
Hybrid	Any combination (e.g. base fee plus per bed)	Built into the engine

Pricing engine capabilities

- Create plans: name, model, rate, included modules, limits, billing cycle
- Custom deals per customer (special rates, discounts, free months) for the sales team
- Free or discounted periods for pilot customers
- Proration when a customer changes plan
- Currency and tax handled with pricing (for other countries later)

Core subscription features

- Self sign-up and a free trial (length to be decided)
- Onboarding: organization details, doctors, rates, and import of existing patient data from Excel
- Features switched on/off by plan; limits on users, branches or beds
- Monthly/yearly billing through Razorpay, auto-renewal, reminders on failed payments
- **Finmeyo's own GST invoice** on every subscription (18% GST)
- Super-admin panel for Finmeyo's team: all customers, plans, usage, support tickets
- In-app support and ticketing

SaaS operations (Finmeyo internal)

New customer (tenant) setup

- From sign-up to live in about 10 minutes through a setup wizard: organization, branches, departments, doctors, rates, print templates
- Selecting a specialty loads its forms, services and tests automatically (e.g. a "dental clinic starter pack")
- A sandbox with demo data so customers can explore safely

Data migration from existing software

- Excel/CSV import with templates for patients, doctors, services, medicine stock and past bills
- Pre-import error report (duplicate mobile numbers, missing names)
- Assisted migration by Finmeyo's team for large customers (paid service)

Feature flags and plans

- On/off switches for every module and major feature, by plan
- Enable a feature for a single customer (beta or special deal)
- Limits: users, doctors, branches, beds, WhatsApp messages, storage
- Crossing a limit prompts an upgrade; work is never blocked

Usage and billing

- Per-customer usage: active users, appointments, messages, storage
- Add-on billing for extra usage (e.g. WhatsApp message packs)
- Commission tracking for partners and resellers

White-label (large customers and resellers)

- Own logo, colours and name
- Own domain (e.g. `app.cityhospital.in`)
- Patient app published under the customer's name on Play Store/App Store (premium add-on)

Support

- In-app chat and tickets, help articles and videos
- Support staff can view a customer's screen only with the customer's permission (time-limited, audit-logged)
- Customer health score: usage levels and churn risk

Releases

- All customers on one version; updates without downtime
- New features roll out to a few customers first, then to everyone
- In-app "What's new" notes

When a customer leaves

- Full data export (Excel, PDF reports, documents)
- Data deleted 90 days after account closure, with prior notice

Security, privacy, hosting and performance

Patient medical data is the most sensitive data there is, so security is built in from day one, not added later.

Security and privacy

- All data encrypted in transit (HTTPS) and at rest
- Data stored on servers in India (data residency)
- Login with password plus OTP; two-factor authentication mandatory for admins and doctors
- Role-based access and a complete audit log
- **DPDP Act 2023:** patient consent, right to access and erase data, breach notification
- Strict tenant isolation: no customer can ever see another's data
- Automatic logout on inactivity

Backup and uptime

- Automatic daily backups retained for at least 30 days
- Uptime target of 99.5% or better on shared plans; 99.9% contractual SLA on enterprise plans
- Customers can export all their data (Excel/PDF) if they leave

Performance

- Patient search and bill generation in under 2 seconds
- 50+ concurrent users per customer during peak hours

Hosting: Hostinger VPS

The whole platform runs on Hostinger VPS. A VPS leaves server management to Finmeyo (no managed services as on AWS), so these rules apply from day one:

- **Location:** the VPS data centre must be in India (DPDP and patient data); confirm the location before buying the plan
- **Docker deployment:** the whole app runs in containers, so it can move to a larger server or AWS/Azure without code changes
- **For the pilot:** one VPS (app and database) is enough; as customers grow, the database moves to its own VPS, then multiple app servers behind a load balancer

- **Backups:** encrypted daily database backups stored **off the VPS** (object storage or another server); Hostinger snapshots alone are not enough
- **Restore tests:** restore from backup every month to prove the data comes back
- **Security:** firewall (only required ports open), key-only SSH, automatic security updates, fail2ban, HTTPS (Let's Encrypt)
- **Monitoring:** instant alerts (WhatsApp/email) for downtime, full disks, high CPU/RAM
- **Staging server:** a small separate VPS where every update is tested before going live
- **Files:** reports, scans and documents in object storage, not on the VPS disk

ABDM / ABHA (national digital health mission)

- Create or link a patient's ABHA ID
- Share health records over the ABDM network with patient consent
- Included in Release 1.0 (ABDM M1–M3), as it is required for insurance claims over NHCX

Interoperability standards

- **HL7 FHIR R4** for health records exchange, as required by ABDM (ABDM FHIR profiles for prescriptions, diagnostic reports, discharge summaries, OP consult notes)
- **HL7 v2 / ASTM** for lab analyzers; **DICOM** and DICOM MWL for imaging
- **Clinical coding:** ICD-10 for diagnoses, SNOMED CT for clinical terms, LOINC for lab tests, standard drug codes for medicines
- Data stored in a structured, coded form from day one so reporting, insurance claims and exchange work without rework

Disaster recovery and business continuity

- Targets: **RPO** (maximum data loss) of 15 minutes through continuous database log shipping; **RTO** (time to restore service) of 4 hours
- Standby copy of the database in a second location, kept in sync, ready to take over
- Documented recovery runbook, tested by a full recovery drill every quarter
- **Downtime procedures** for hospitals: printable downtime forms, read-only offline copy of the day's appointments, admitted patients and medication charts at each nursing station, and back-entry of records once systems return
- Status page and WhatsApp alerts to customers during incidents

Enterprise platform requirements

These requirements apply across all modules. They are what hospital chains, their IT and security teams, and their legal departments check before signing, and they separate an enterprise product from a clinic tool.

1. Identity and access

- **Single sign-on:** SAML 2.0 and OpenID Connect with Microsoft Entra ID (Azure AD), Google Workspace, Okta and others
- **SCIM provisioning:** users created, updated and deactivated automatically from the customer's directory
- **Shared workstation mode:** fast user switching at nursing stations and front desks by badge tap or PIN, with automatic lock on inactivity
- Password policies per customer (length, expiry, reuse), MFA enforcement by role
- IP allowlisting and login restriction by network or time of day
- Active session list per user, with remote sign-out
- **Break-the-glass access:** in an emergency, a clinician can open a record outside their normal access after stating a reason; every such access is logged, alerted and reviewed
- Fine-grained permissions down to field level (e.g. hide HIV status or psychiatric notes from most roles)

2. Organization hierarchy for hospital chains

- Hierarchy: group → region → hospital (legal entity) → branch → department → unit
- Multiple legal entities under one group, each with its own GSTIN, books and invoice series
- Inter-company transactions (e.g. lab tests done by one entity for another) with automatic inter-company accounting
- Consolidated reporting at any level of the hierarchy
- Group-level masters and policies (rate lists, formularies, templates) that branches inherit and cannot change, with controlled local overrides
- Users with access scoped to any node (e.g. a regional manager sees only their region)

3. Deployment options

Option	For	Notes
Shared SaaS	Clinics and mid-size hospitals	Default; Hostinger VPS infrastructure at launch
Dedicated instance	Large hospitals and chains	Separate servers and database for one customer, managed by Finmeyo
Private cloud	Customers with cloud preferences	Deployed in the customer's AWS/Azure/GCP account

Option	For	Notes
On-premise	Government and security-sensitive hospitals	Installed on the customer's servers; updates shipped as versioned packages

One codebase serves every option; containerised deployment (Docker/Kubernetes) makes this possible. Infrastructure moves to a larger cloud provider when customer scale requires it.

4. Security certifications and compliance

- **CERT-In directions (mandatory in India):** report cyber incidents to CERT-In within 6 hours; keep system logs for 180 days in India; synchronise clocks with NTP servers
- **DPDP Act 2023:** consent records, data principal requests (access, correction, erasure) handled in a tracked workflow, breach notification to the Data Protection Board and affected patients, a named Data Protection Officer
- **Security testing:** annual third-party VAPT and before every major release; findings tracked to closure
- **Certifications:** ISO 27001 (information security) and SOC 2 Type II as targets within the first year of launch; ABDM security requirements for integration
- **International readiness:** HIPAA and GDPR controls designed in for later markets
- **Data governance:** masked or synthetic data in non-production environments; field-level encryption for sensitive data (HIV, psychiatry, MTP); data classification; secrets management; least-privilege access for Finmeyo staff with approval and logs
- Security policies documented: access control, incident response, vendor risk, secure development

5. SLA and support

- Contractual uptime SLA of 99.9% for enterprise plans, with service credits when missed
- Severity-based response and resolution targets:

Severity	Example	Response	Resolution target
S1 Critical	System down, billing or EMR unusable	15 minutes	4 hours
S2 High	A module down, major feature broken	1 hour	1 business day

Severity	Example	Response	Resolution target
S3 Medium	Feature partly working, workaround exists	4 hours	5 business days
S4 Low	Question, minor issue	1 business day	Next release

- 24x7 support for S1/S2 issues; business-hours support for others
- Dedicated customer success manager for enterprise customers, with quarterly business reviews
- Public status page and incident post-mortems shared with affected customers

6. Customer sandboxes and change control

- A sandbox/UAT environment per enterprise customer, refreshed from production with masked data
- Configuration promotion: masters, forms, templates, workflows and roles moved from sandbox to production with a change log and approval
- Release preview: enterprise customers get new versions in their sandbox two weeks before production
- Scheduled update windows and the option to defer non-critical updates for a limited period
- Configuration version history with rollback

7. Integration platform

- Built-in interface engine for HL7 v2, FHIR, DICOM and custom formats, with message mapping, queueing, retries and a message log
- Connectors to existing hospital systems during migration (legacy HIS, LIS, PACS, ERP)
- Versioned public APIs (v1, v2…) with deprecation notices of at least 12 months
- Developer portal: API documentation, sandbox keys, SDKs, webhook testing
- OAuth 2.0 for third-party apps, with per-app scopes the customer approves
- Later: an app marketplace where partners publish add-ons for customers to install

8. Engineering quality, scale and observability

- Automated test suites (unit, integration, end-to-end) run on every change; no release without passing tests

- CI/CD pipeline with code review, static analysis and dependency vulnerability scanning
- Load testing to the profile of a 500-bed hospital at peak (e.g. morning OPD rush, end-of-month billing), repeated before major releases
- Horizontal scaling of application servers; database read replicas for reports
- Per-tenant isolation of heavy workloads so one customer's large report cannot slow others
- Observability: application performance monitoring, centralised logs (kept 180 days for CERT-In), metrics per customer, alerting with on-call rotation
- Feature flags and staged rollouts (module-level and customer-level)

9. Implementation and training

- Standard implementation methodology: kickoff → requirement mapping → configuration → data migration → training → UAT → go-live → 30-day hypercare → handover to support
- Implementation plan templates by customer size (clinic: days; mid-size hospital: weeks; large chain: phased by branch)
- Go-live readiness checklist and sign-off
- Training academy: role-wise video courses, quizzes and certificates for staff (front desk, nurse, doctor, billing, TPA, store)
- In-app guided walkthroughs and contextual help
- Train-the-trainer programme for large customers
- Certified implementation partners to scale delivery beyond Finmeyo's own team

10. Legal and contractual documents

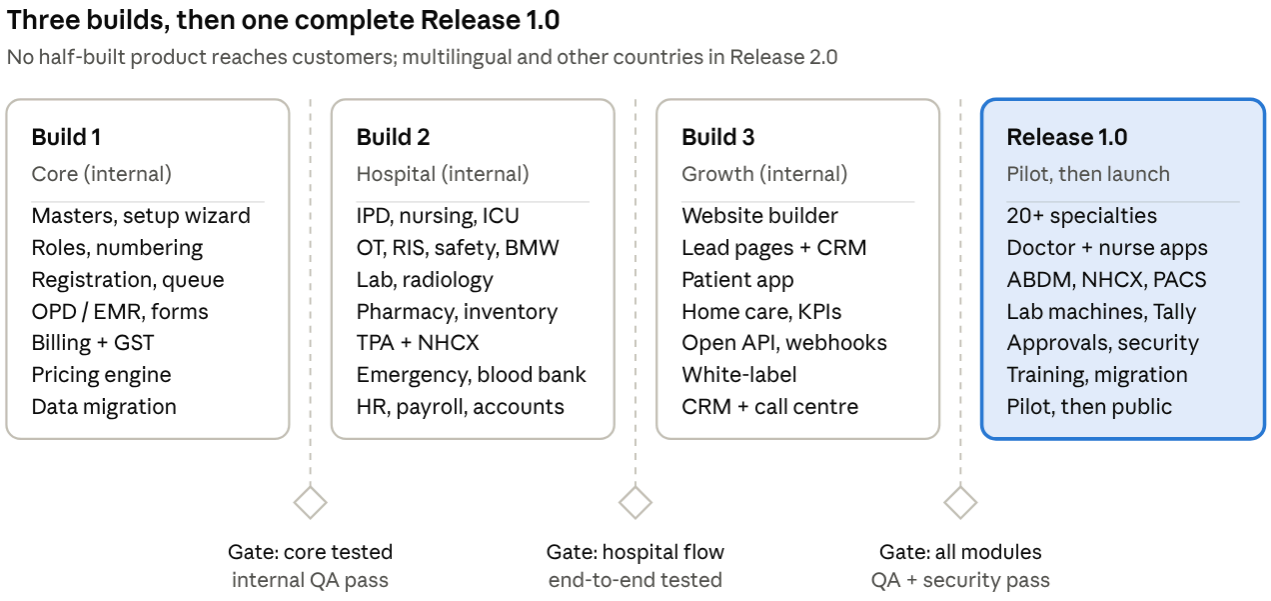
Document	Purpose
Master Services Agreement (MSA)	Overall commercial and legal terms
Order form	Plan, modules, price, term for each customer
Service Level Agreement (SLA)	Uptime, support response, service credits
Data Processing Agreement (DPA)	Finmeyo as data processor of patient data under DPDP; security obligations, sub-processors, breach handling
Privacy policy and terms of use	For the platform, patient app and websites
Business Associate Agreement (BAA)	For US customers later (HIPAA)

Document	Purpose
Implementation statement of work	Scope, timelines and responsibilities for go-live
Partner/reseller agreement	For implementation partners and resellers

All documents to be drafted and reviewed by a lawyer before the first pilot contract.

Roadmap

Development runs in three internal builds, but customers only ever see the complete Release 1.0. After all three builds are finished and tested, the product runs with 2–3 pilot hospitals, then launches publicly. Claude writes the code, so development is fast; government and vendor approvals (ABDM, NHCX, WhatsApp, Razorpay KYC, DLT) and pilot testing still take calendar time, so they start on day one.



roadmap · 3 builds, then Release 1.0

Each gate means the next step starts only when the previous one is fully tested. Pilot feedback sets the pricing. The length of each build depends on team and budget.

When the configuration features arrive: masters, role builder, numbering, print templates, form builder, workflow settings, approvals, setup wizard and pricing engine are all built in **Build 1**, because without them the product cannot be generic. Clinical safety and nursing, infection control and compliance, and RIS are part of Build 2. Website builder, lead pages, CRM and call centre, KPIs and analytics, health check-ups, home care, pharmacy delivery, facility services, patient app, open API and white-label come in **Build 3**.

Release 1.0 launches only when all three builds are complete. Multilingual support, other countries, organ transplant (THOA), teaching-hospital features, clinical research and AI features (voice notes, coding suggestions) follow in **Release 2.0**.

Open items

Decisions already made are listed under Product decisions. These remain open:

- [ ] Per-clinic subscription fee (decide after the pilot)
- [ ] Length of the free trial
- [ ] App name and branding (later)
- [ ] Which 3–5 hospitals/clinics in Delhi NCR will pilot
- [ ] Start registrations in Finmeyo's name: ABDM, NHCX, WhatsApp Business, Razorpay KYC, SMS DLT
- [ ] Update Finmeyo's MOA objects clause and GST registration to cover software/SaaS
- [ ] Choose the Hostinger VPS plan, confirm an India data centre, and pick off-VPS backup storage
- [ ] Get MSA, SLA, DPA, privacy policy and terms drafted by a lawyer before the first pilot contract
- [ ] Appoint a Data Protection Officer and set up the CERT-In incident reporting process
- [ ] Plan the first VAPT before pilot go-live, and the ISO 27001 / SOC 2 roadmap

Change log

Version	Date	Change
1.0	5 Nov 2026	First complete, frozen requirements: 26 modules and enterprise platform requirements, all in Release 1.0

Next documents

Building starts from these, in this order: technical architecture, data model, build plan, user stories with acceptance criteria per module, UI/UX design, and the business plan for the Delhi NCR pilot.